Customer Churn Analysis Dashboard

1. Project Overview

Customer churn is a critical problem for telecom companies because losing customers directly impacts revenue and growth. This project focuses on analyzing customer data to identify patterns and factors that influence customer churn.

Using a telecom dataset, an interactive dashboard was developed in **Microsoft Power BI** to explore customer behavior, service usage, and churn patterns. The dashboard helps visualize key insights that can support business decisions aimed at improving customer retention.

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2. Problem Statement

Telecommunication companies experience significant customer churn due to service dissatisfaction, pricing, and competition. Identifying customers who are likely to churn and understanding the key drivers behind churn is essential for designing retention strategies.

The goal of this project is to:

- Analyze customer demographics and service usage
- Identify patterns associated with customer churn
- Provide actionable insights to reduce churn

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3. Dataset Description

The dataset contains **7043 customer records** and **23 attributes** describing customer demographics, services subscribed, billing details, and churn status.

Key Variables

Customer Information

- Customer ID
- Gender
- Senior Citizen
- Partner
- Dependents
- Tenure (number of months with the company)

Services Used

- Phone Service
- Multiple Lines
- Internet Service
- Online Security
- Online Backup
- Device Protection
- Tech Support
- Streaming TV
- Streaming Movies

Billing Information

- Contract Type
- Paperless Billing
- Payment Method
- Monthly Charges
- Total Charges

Support Interaction

- Number of Administrative Tickets
- Number of Technical Tickets

Target Variable

- Churn (Yes / No)

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4. Data Preparation

The dataset was cleaned and prepared before building the dashboard.

Key steps included:

- Checking for missing or inconsistent values
- Converting categorical fields into appropriate formats
- Verifying numerical fields such as **MonthlyCharges** and **TotalCharges**
- Preparing the churn variable as the target metric for analysis

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5. Dashboard Features

The Power BI dashboard provides interactive visualizations to analyze churn patterns.

Key components include:

Key Performance Indicators (KPIs)

- Total Customers
- Total Churned Customers
- Churn Rate
- Average Monthly Charges

Visualizations

The dashboard includes charts analyzing:

- Churn by Gender
- Churn by Contract Type
- Churn by Internet Service
- Churn by Payment Method
- Churn by Tenure
- Churn by Senior Citizen Status
- Service usage vs churn rate

Filters and slicers allow users to dynamically explore customer segments.

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6. Key Insights

Based on the analysis, several important patterns were observed:

- Customers with **month-to-month contracts** show significantly higher churn rates.
- Customers paying through **electronic check** are more likely to churn.
- Customers with **short tenure** tend to churn more frequently.
- Customers without **online security or tech support services** have a higher likelihood of leaving.
- Senior citizens show slightly higher churn rates compared to other groups.

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7. Business Recommendations

Based on the analysis, the following strategies could help reduce churn:

1. Encourage customers to switch from **month-to-month contracts to long-term contracts**.
2. Offer incentives or discounts for customers with **short tenure** to improve retention.
3. Promote **value-added services** such as online security and technical support.
4. Improve customer service support for users generating high numbers of support tickets.
5. Provide targeted retention offers to high-risk customer segments.

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8. Tools Used

- Microsoft Power BI – Data visualization and dashboard development
- Microsoft Excel – Dataset storage and preparation

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9. Project Outcome

This dashboard demonstrates how data visualization and analytics can help telecom companies identify churn patterns and develop strategies to improve customer retention and overall business performance.